

Hollow Concrete Blocks: Production and Use in EQ Affected Districts

14th December 2016

NRA Conference Hall

Information Gathered

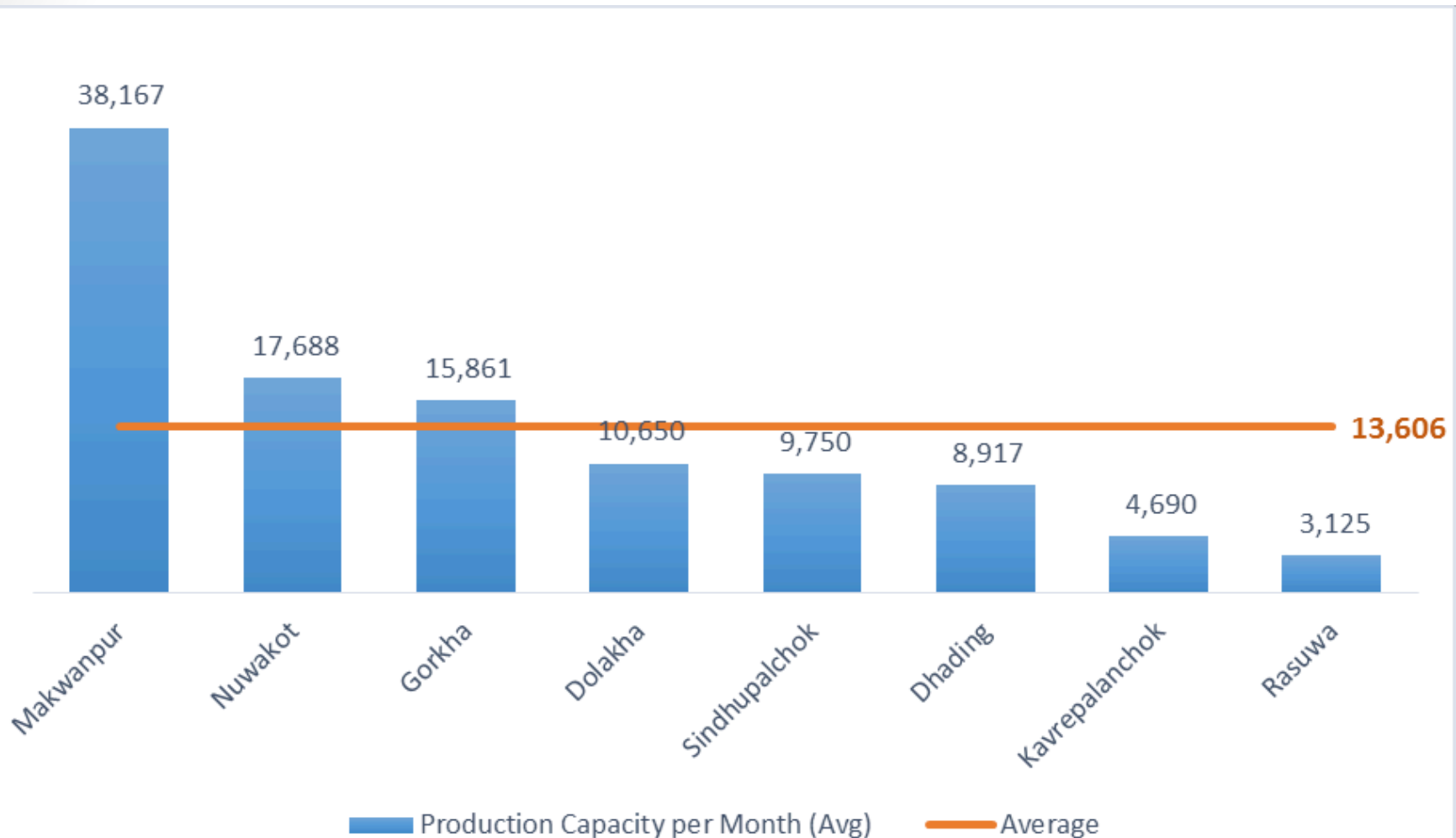
- From producers:
 - Date of casting
 - Proportion of mix,
 - Curing time and method,
 - Brand of cement used,
 - Cost per block,
 - What is production capacity? (per day or week or month)
 - Type of mould used, and
 - Contact details.

Information Gathered – Contd.

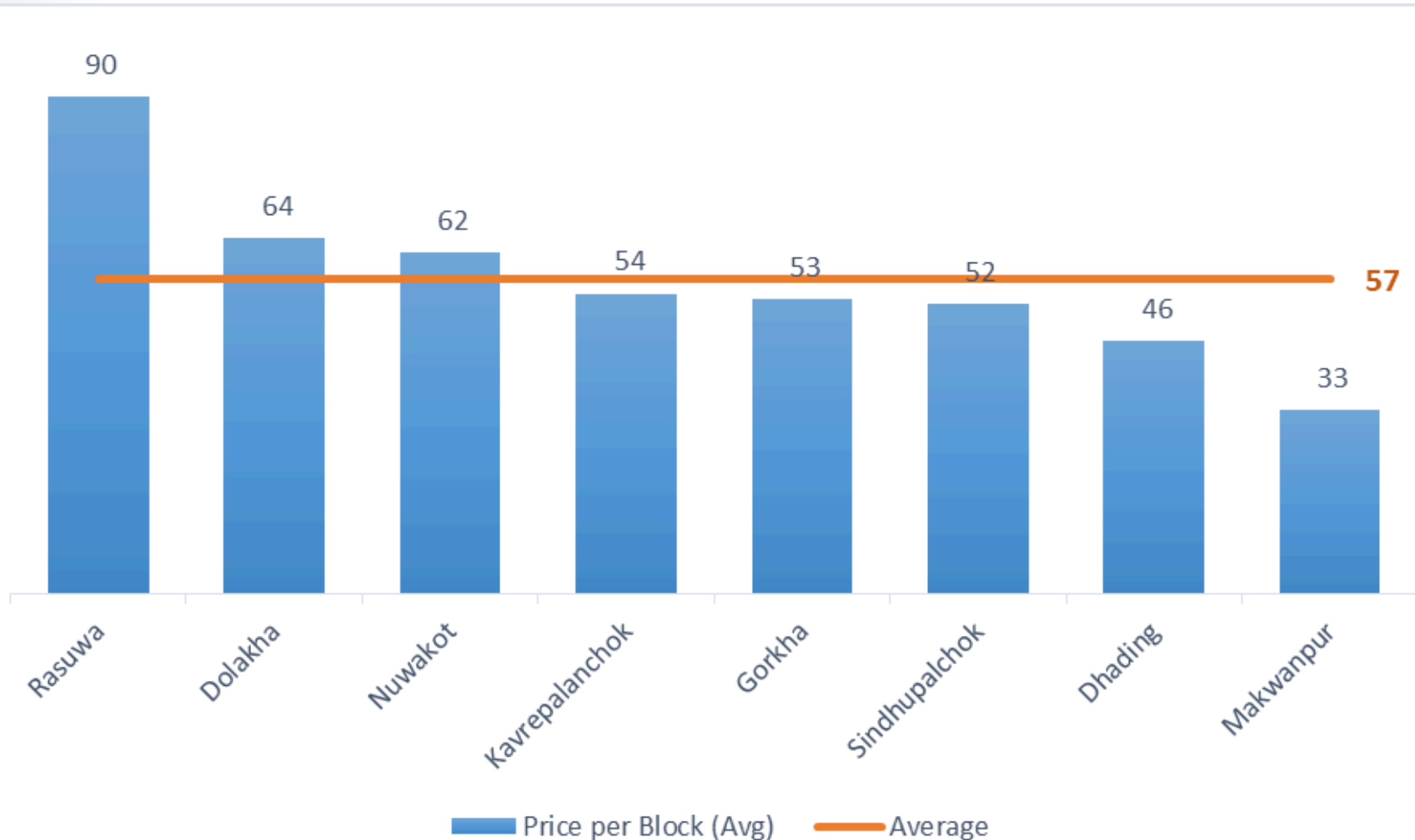
- From households:
 - Where did you buy the blocks from?
 - How much did the blocks cost?
 - Why did you choose to use blocks?
 - Did you receive any orientation / information on 'BBS'? If so, how did you incorporate this into the construction of your house?
 - Had the mason / person who constructed the house worked with blocks previously?
 - Did you seek information on using blocks from anywhere? If yes, where and what information were they given?
 - Are you eligible for the GoN housing grant?

BLOCK PRODUCTION

Production Capacity



Price per Block



Mix Ratio

- Varied a lot; 1:9:9, 1:3:4, 1:3:6, 1:10:4, 1:4:4, 1:5:7
- One producer in Dhading stated that they consult an engineer to check the mix
- In some cases producers were very unhappy about being asked this information and were unwilling to share





Curing

- Majority curing for one week, some for less than that (3-4 days), some for slightly longer, and very small number curing blocks for between 21 and 28 days
- Several producers indicated that they knew they should not sell the blocks just 7 days after production, but that they are selling at this point due to the high demand
- Curing involves either pouring water over the blocks two to four times per day, or submersion in pond









Testing

- 10 blocks have been tested, breaking strength was as follows:
 - 1 – 36.67 kg/cm²
 - 2 – 13.69 kg/cm²
 - 3 – 15.27 kg/cm²
 - 4 – 12.74 kg/cm²
 - 5 – 40.84 kg/cm²
 - 6 – 57.80 kg/cm²
 - 7 – 21.12 kg/cm²
 - 8 – 15.70 kg/cm²
 - 9 – 38.63 kg/cm²
 - 10 – 14.53 kg/cm²
- IS 2185 Part 1:
- Min. compressive strength of blocks range between 28 kg/cm² to 122.37 kg/cm² for different grades





USE OF BLOCKS

Key Findings

- None of the households interviewed had received information regarding construction with blocks
- Some had received some information about BBS / EQ resistant construction elements
- Reasons for selecting blocks were:
 - Requires less water and there is a scarcity of water
 - Cheaper than other materials
 - Can build more quickly
 - Saw others building with blocks
 - Good stone is not available
 - Stone is hard to store
 - Bricks are expensive to import to the area
- Only one household reported that the mason they hired had not worked with concrete blocks before



























Next Steps

- Continue to document use of blocks, identifying common issues / non-compliances
- Include information on concrete blocks in Technical Assistance package
- Provide TA to **all households**
- Continue documentation and testing of block production in districts
- Develop and implement packages of support for block producers